BiomationScripter: A Python Library for Developing Synthetic Biology Automation Protocols

-- Methods

Preparation of Level 0 Loop Assembly Parts

Level 0 phytobricks were created by Polymerase Chain Reaction (PCR) amplification from a DNA template. Primers were designed to add 5' and 3' ends to allow for assembly into a Level 0 phytobrick acceptor plasmid (*pSB1C00*). PCRs were performed manually. Reagents and volumes used can be found in Table 1, with thermocycling conditions defined in Table 2. All level 0 parts created, along with a list of primers used, can be found in Table 3. PCR fragments were analysed by gel electrophoresis to confirm correct sizing. All PCR fragments were assembled into *pSB1C00* manually via loop assembly. Reactions were prepared as described in Table 4, and then subjected to thermocycling as shown in Table 5. 5 µL of reaction mixture was transformed into *E. coli* DH5α and plated onto LB agar plates with chloramphenicol (50 µg/mL) antibiotic. A number of non-red colonies (the *pSB1C00* plasmid contains a red fluorescent protein expression unit in the insertion site) per transformation were selected and subjected to plasmid purification (New England Biolabs; Monarch Miniprep Kit; as per manufacturer's instructions). The sequence of level 0 parts in were confirmed via Sanger sequencing (Eurofins). Sequencing samples were prepared by combining approximately 500 ng of plasmid DNA with 2.5 µL of either 10 µM forward primer (VF2: TGCCACCTGACGTCTAAGAA) or 10 µM reverse primer (VR: ATTACCGCCTTTGAGTGAGC) and making up to 10 µL with nuclease-free water.

Automated GoldenGate Assembly

Four level 1 phytobrick assemblies (Table 6) were prepared using the Echo525 (Labcyte) and the BiomationScripter Loop assembly Template from the level 0 parts created as described above. Reactions were performed at 5 µL final volume and prepared in a 384-well, v-bottom PCR plate. Proportions of reagents per reaction are shown in Table 7. Nuclease-free water was transferred from a 6RES plate (Labcyte). Level 0 DNA parts, *pOdd* plasmids, and buffer were transferred from 384PP plates (Labcyte). All other reagents were transferred from 384LDV (Labcyte) plates. The PCR plate had a foil seal applied and was vortexed briefly before and centrifugation for approximately 10 seconds before being thermocycled according to Table 5. For analsysis, 5 µL of assembly reaction was used to transform *E. coli* DH5α cells, and colonies were selected for plasmid purification. Plasmids were subjected to manual restriction digest with BsaI-HF (20,000 units/mL; New England Biolabs) according to manufacturer's instructions, and fragments were analysed using the TapeStation 4200 (Agilent) and the D5000 Tape Kit (Agilent), according to the manufacturer's instructions. A picture of the gel was exported from the TapeStation Analysis Software (v4.1.1), as seen in the main text (Figure 2aii). All assemblies were also verified by Sanger Sequencing (Eurofins).

Automated Plate Reader Calibration Plate Preparation

In a black 96-well plate with flat, clear bottomed wells, 1 in 2 serial dilutions of the standard calibrants were prepared to a final volume of 100 µL across 11 wells, with a twelfth well containing no calibrant. Each serial dilution was replicated twice. The calibrants used were as follows: fluorescein (Merck) in

Phosphate-Buffered Saline (PBS), sulforhodamine 101 (Merck) in PBS, cascade blue (ThermoFisher Scientific) in water, and 950 nm monodisperse silica microspheres (Nanocym) in water. The first concentration of each calibrant in the serial dilutions was 10 µM for fluorescein, 2 µM for sulforhodamine 101, 10 µM cascade blue, and 3e9 particles/mL for the microspheres. The serial dilutions were prepared using the BiomationScripter OTProto Standard iGEM Calibration Template. Fluorescence and absorbance measurements were made using a Clariostar Plus (BMG Labtech).

Table 1: PCR Reaction Mixture Composition

Reagent	Amount
Q5-HF Master Mix (10x)	12.5 µL
Forward Primer (10 µM)	1.25 µL
Reverse Primer (10 µM)	1.25 µL
Template DNA	~800 ng
Nuclease-Free Water	Up to 25 µL

Table 2: PCR Thermocycling Conditions

Step	Number of Cycles	Temperature (°C)	Time (Seconds)
1	1	98	30
2	35	98	10
		60	30
		72	40
3	1	72	120
4	1	4	Inf

Table 3: PCR Reactions: Shown here are PCR reactions performed, with DNA templates and primers used.

PCR Fragment Name	Template DNA	Forward Primer	Reverse Primer
J23100_B0034_AC	PBLInd100-EL222 (Table 8)	ATCAgctcttcATCGggagtctTTGACGGCTAGCTCAGTCCT	CTAGCCTAGAGAAAGAGGAGAAATACTAGaatgCGAGTgaagagcttgc
PBLInd100_B0034_AC	PBLInd100-EL222 (Table 8)	ATCAgctcttcATCGggagGGTAGCCTTTAGTCCATGtt	taaaTATGTCTAGAGAAAGAGGAGAAATACTAGaatgCGAGTgaagagcttgc
EL222_CE	PBLInd100-EL222 (Table 8)	ATCAgctcttcATCGaatgTTGGATATGGGACAAGATCG	CCGTCGAAGCCGGAATCTAAgcttCGAGTgaagagcttgc
mCherry_CE	Default Processor + mCherry Module (Table 8)	ATCAgctcttcATCGaATGGTGAGCAAGGGCGAGGAGG	CATGGACGAGCTGTACAAGTAAgcttCGAGTgaagagcttgc
B0015_EF	PBLRep-EL222 (Table 8)	ATCAgctcttcATCGgcttaggatctccaggcatcaaataaaac	tcgggtgggcctttctgcgtttatacgctCGAGTgaagagcttgc

Table 4: Level 0 Loop Assembly: (*) For each reaction, DNA backbone and DNA parts were added in a 1:3 ratio (backbone:part)

Reagent	Amount (µL)
SapI-HF (NEB)	1.25
T4 Ligase Buffer (NEB)	5
T4 Ligase	1.25
DNA Backbone (10 fmol/µL)*	2.5
DNA Parts (10 fmol/µL)*	Various*
Nuclease-Free Water	Up to 50

Table 5: Loop Assembly Thermocycling Conditions

Step	Number of Cycles	Temperature (°C)	Time (Minutes)
1	30	37	3
		16	4
2	1	50	5
3	1	80	10
4	1	4	Inf

Table 6: DNA Constructs Assembled via Automated fLoop Assembly: Shown here are assembled DNA constructs assembled with Loop assembly using the EchoProto template.

Construct Name	Plasmid Backbone	Antibiotic Selection	DNA Parts	Link to Sequence
pOdd1: PBLInd-mCherry	pOdd1 (Table 8)	Kanamycin	• PBLInd100_B0034_AC (Table 3) • mCherry_CE (Table 3) • B0015_EF (Table 3)	benchling.com/s/seq-GZuvozOBcFWx75SNatvN?m=slm-EvXvnAGnUXFmK3migJ0K
pOdd1: J23100-mCherry	pOdd1 (Table 8)	Kanamycin	• J23100_B0034_AC (Table 3) • mCherry_CE (Table 3) • B0015_EF (Table 3)	benchling.com/s/seq-NA9sMgwo22g9MChF1Qmg?m=slm-i7QlWkXxptUJztdUqcnZ
pOdd2: PBLInd-EL222	pOdd2 (Table 8)	Kanamycin	• PBLInd100_B0034_AC (Table 3) • EL222_CE (Table 3) • B0015_EF (Table 3)	benchling.com/s/seq-5y5IyNioaWE29xJpkXwm?m=slm-1eo8OYEEB4Ry2k3BZZV7
pOdd2: J23100-EL222	pOdd2 (Table 8)	Kanamycin	• J23100_B0034_AC (Table 3) • EL222_CE (Table 3) • B0015_EF (Table 3)	benchling.com/s/seq-iiUaTlm97fh753jBtYfg?m=slm-uekx5RVwp8PVBTmyJKHf

Table 7: EchoProto Level 1 Loop Assembly: (*) For each reaction, DNA backbone and DNA parts were added in a 1:1 and 1:2 ratio (backbone:part)

Reagent	Amount (µL)
BsaI-HF (NEB)	0.125
T4 Ligase Buffer (NEB)	0.5
T4 Ligase	0.125
DNA Backbone (10 fmol/µL)*	0.25
DNA Parts (10 fmol/µL)*	Various*
Nuclease-Free Water	Up to 5

Table 8: DNA Obtained Outside this Study: Listed here are all DNA parts or constructs with did not originate within this study.

Name	Plasmid Backbone	Antibiotic Selection	Source
PBLRep-EL222	----	Kanamycin	Gift from Chueh Loo Poh [1]
PBLInd100-EL222	----	Kanamycin	Gift from Chueh Loo Poh [1]
pOdd1	pOdd1	Kanamycin	iGEM 2017 Distribution Kit
pOdd2	pOdd2	Kanamycin	iGEM 2017 Distribution Kit
pSB1C00	pSB1C00	Chloramphenicol	iGEM 2017 Distribution Kit
Default Processor + mCherry Module	pSB1C3	Chloramphenicol	Assembled by authors - benchling.com/s/seq-GgM5DP5lxw2ixtjvjkjo

[1] P. Jayaraman, K. Devarajan, T. K. Chua, H. Zhang, E. Gunawan, and C. L. Poh, 'Blue light-mediated transcriptional activation and repression of gene expression in bacteria.', *Nucleic Acids Res*, vol. 44, no. 14, pp. 6994–7005, 2016, doi: 10.1093/nar/gkw548.